

EE 508P: Practicum on Electric Drives

Course Code	EE 508P
Course Name	Practicum on Electric Drives
L-T-P-C	0-0-3-2
Prerequisites	EE 201 and EE 201P- Electromechanics or Equivalent, EE 309 and EE 309P- Power Electronics or Equivalent
Intended for	Final year BTech Electrical Engineering (EE), M.Tech in Power Electronics and Drives (PED)
Distribution	Core for 1 year M.Tech. (PED), Elective for other PG and BTech Final year EE
Approval	17th BoA (May 2017); 13th Senate (Jun 2017)

This is a laboratory course with 3-hour sessions per week. Following is the list of course modules and experiments.

- **Introduction to Drives** [6 hours]
 - Calculation of moment of inertia of drive+ load system
 - Verification of Fundamental Drive Equation
 - . Study of types of loads and their torque-speed characteristics
- **DC Drives** [12 hours]
 - Characteristics of different types of DC motors (series, shunt, separately excited)
 - Speed control (acceleration, braking) of DC motors (Rectifier fed, chopper fed)
 - Dynamic control of DC drives (closed loop controller design, simulation and validation)
 - Regenerative braking in closed loop DC drives
 - Closed loop controller design and digital implementation of separately excited DC motor drive
 - Introduction to field weakening mode operation and effect on dynamic performance
- **Induction Motor Drives** [15 hours]
 - No-load/blocked rotor test on induction motor
 - Plotting speed-torque characteristics of an induction motor
 - Stator voltage control of induction motor (fed with VSI)
 - V/f control of induction motor (fed with VSI)
 - Slip speed control of induction motor (fed with VSI)
 - Rotor resistance control of slip-ring induction motor
 - Study of three-phase self-excited induction generator (SEIG)
- **Synchronous Motor Drives** [6 hours]
 - Driving the motor from a variable voltage and frequency supply
 - Study of V-curves
- **Speed control of Special machines** [3 hours]
 - BLDC, PMSM etc.

Text Books

Lab. experimental manuals will be provided.

References

1. W. Leonhard, **Control of Electrical Drives**, Springer-Verlag, 2001.
2. Mohan N., Undeland T. M. and Robbins W. P., **Power Electronics- Converters, Applications and Design**, 3rd Edition, Wiley India, 2008.
3. Bose B. K., **Power Electronics and Variable Frequency Drives - Technology and Applications**, IEEE Press, Standard Publisher Distributors. 2001.
4. Rashid M., **Power Electronics - Circuits, Devices and Applications**, 3rd Edition, Pearson Education.
5. Krause, P. C., Wasynczuk, O., Sudhoff, S.D., **Analysis of Electric Machinery and Drive Systems**, Wiley-Interscience.